

Sworup Bhandari

AI & Backend Engineer

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SUMMARY

AI & Backend Engineer who builds production-grade systems end to end, from fine-tuned transformer models to the distributed services that serve them. Architected an AI-assisted security scanner pairing a DistilBERT classifier with a NestJS microservice backend on a 59k-sample dataset, with strengths across Python ML pipelines, TypeScript/Node backends, and real-time full-stack delivery.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Ruby, C/C++, Dart
- **AI / ML:** PyTorch, Transformers (DistilBERT), XGBoost, YOLOv8, ONNX/TorchScript, feature engineering, model evaluation
- **Backend:** NestJS, FastAPI, Express.js, Ruby on Rails, REST
- **Data & Infra:** PostgreSQL, Redis, BullMQ, Docker, Docker Compose, microservices, async job queues
- **Frontend:** React, Next.js, Flutter
- **Security:** XSS detection, context-aware payload generation, fuzzing, WAF analysis, headless-browser verification (Playwright)
- **Practices:** system architecture, MVC, auth (JWT/API keys), e2e/unit testing, Git, Linux

FLAGSHIP PROJECT

RedSentinel: AI-Assisted XSS Vulnerability Scanner Final-year Capstone, Team of 3 | [Code](#) | [Report](#) | [Demo](#)

Lead architect; owned the NestJS orchestration core and the end-to-end ML pipeline. Teammates built the fuzzer and obfuscation engines.

- Architected a polyglot microservice system: a **TypeScript/NestJS** core orchestrating three **Python/FastAPI** AI services over a **Redis/BullMQ** job queue, with **PostgreSQL** persistence and a real-time **Next.js** dashboard (Socket.IO).
- Fine-tuned a **DistilBERT** multi-task classifier to label an injection point's context and severity, replacing brittle signature matching, and reached near-perfect accuracy on an **8,869-sample** held-out split of a deterministic, well-structured label space.
- Built the full ML pipeline on a **59,122-entry** curated XSS dataset: data curation, custom tokenizer training, multi-task training with checkpointing, evaluation, and an inference API with ONNX/TorchScript export and confidence calibration.
- Co-developed an **XGBoost** payload ranker (35 engineered features) to prioritize payloads by predicted effectiveness, with a heuristic fallback when the trained artifact is unavailable.
- Designed the NestJS core: scan lifecycle, crawler, BullMQ queue processor, audit logging, JWT/API-key auth, WebSocket streaming, and HTML/JSON/PDF report generation, backed by unit and e2e test suites.

PROJECTS

Community Connect

OSM Hackathon | Express.js, Flutter

- Developed a full-stack event-coordination platform with REST APIs for real-time GPS-based attendance validation and event tracking.

XSS Payload Generator — Deep Learning Payload Synthesis

| Python, TensorFlow/Keras

- Designed and trained a decoder-only Transformer model from scratch to generate context-aware XSS payloads, including custom multi-head attention blocks with causal masking

Travel Buddy — Social Travel Planning Platform

| React, Ruby on Rails, PostgreSQL

- Built a full-stack social travel platform featuring trip planning with participant approval workflows, a social feed, friendship/connection management, and PDF itinerary uploads.
- Implemented token-based authentication (Devise Token Auth), real-time chat via Rails Action Cable and interactive trip maps with Leaflet

Note App

| React, Ruby on Rails, PostgreSQL

- Built a full-stack note-taking app with secure user authentication, enabling users to create, organize, and manage personal notes through a responsive UI with Added PDF export

EDUCATION

Institute of Engineering, Tribhuvan University

2022 to 2026

Bachelor's in Computer Engineering, Pashchimanchal Campus, Pokhara